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# =====
# Facial Recognition - Levels of Computer Vision
# Google Colab Compatible
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt

# -----
# Step 1: Low-Level Computer Vision
# Create a synthetic face image
# -----

img = np.zeros((300, 300), dtype=np.uint8)

# Face
cv2.circle(img, (150,150), 100, 200, -1)

# Eyes
cv2.circle(img, (115,120), 10, 0, -1)
cv2.circle(img, (185,120), 10, 0, -1)

# Mouth
cv2.ellipse(img, (150,185), (40,20), 0, 0, 180, 0, 3)

# Noise Reduction (Low-Level)
blur = cv2.GaussianBlur(img, (5,5), 0)

# Contrast Enhancement (Low-Level)
enhanced = cv2.equalizeHist(blur)

# -----
# Step 2: Mid-Level Computer Vision
# Feature Extraction
# -----

edges = cv2.Canny(enhanced, 100, 200)

# -----
# Step 3: High-Level Computer Vision
# Simulated Face Recognition
# -----

result = cv2.cvtColor(enhanced, cv2.COLOR_GRAY2RGB)

cv2.putText(result,
            "Face Recognized",
            (60,25),
            cv2.FONT_HERSHEY_SIMPLEX,
            0.8,
            (255,0,0),
            2)

# -----
# Display Results
# -----

plt.figure(figsize=(14,8))

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plt.subplot(2,2,1)
plt.imshow(img, cmap='gray')
plt.title("Low-Level: Image Acquisition")
plt.axis("off")

plt.subplot(2,2,2)
plt.imshow(enhanced, cmap='gray')
plt.title("Low-Level: Enhanced Image")
plt.axis("off")

plt.subplot(2,2,3)
plt.imshow(edges, cmap='gray')
plt.title("Mid-Level: Feature Extraction")
plt.axis("off")

plt.subplot(2,2,4)
plt.imshow(result)
plt.title("High-Level: Face Recognition")
plt.axis("off")

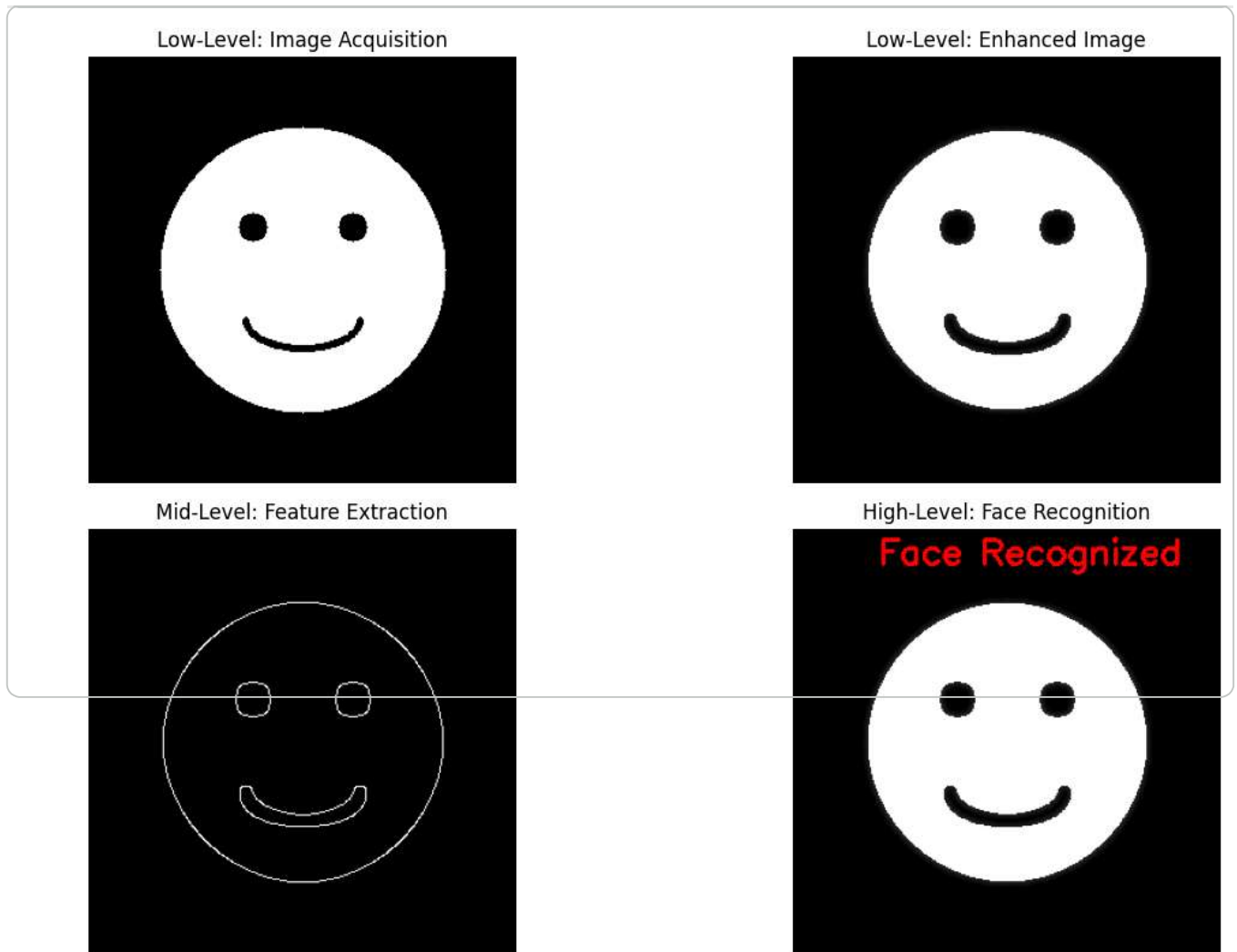
plt.tight_layout()
plt.show()

# -----
# Output
# -----

print("Low-Level Computer Vision:")
print("- Image acquisition")
print("- Noise reduction using Gaussian Blur")
print("- Contrast enhancement using Histogram Equalization\n")

print("Mid-Level Computer Vision:")
print("- Feature extraction using Canny Edge Detection\n")

print("High-Level Computer Vision:")
print("- Simulated face recognition")
print("- Authentication decision: Face Recognized")
```



Low-Level Computer Vision:

- Image acquisition
- Noise reduction using Gaussian Blur
- Contrast enhancement using Histogram Equalization

Mid-Level Computer Vision:

- Feature extraction using Canny Edge Detection

High-Level Computer Vision:

- Simulated face recognition
- Authentication decision: Face Recognized